

Name of the Student	Narra Sai Mounika
Reg. No	192425367
GitHub Link	https://github.com/saimounika1315/MLA0403---Deep-Learning

SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL 1

Case Study

COURSE NAME: DEEP LEARNING
SLOT: A

COURSE CODE: MLA0403

COURSE OUTCOME COVERED

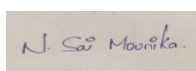
CO3: Analyze and apply convolutional neural network architectures, including convolutional layers, filters, parameter sharing, regularization, AlexNet and ResNet, to develop solutions for image classification and real-world computer vision applications. (L4)

CO Weightage: 25%
Assessment Tool Weightage: 25%

Duration: 90 minutes
Marks: 25

Code of Conduct:

I Narra Sai Mounika (192425367) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Narra Sai Mounika
Name of the student:

Criteria	Conceptual Understanding (2.5)	Linkages (2.5)	Organization (2)	Integration of Literature (2)	Clarity & Presentation (1)	Total (10)
Weightage	25 %	25%	20%	20 %	10 %	100%
Q1						
Q2						
Total						

Signature of the Faculty:

Total Marks:

Case Study Question 1: CNN for Automated Medical Image Classification (10 Marks)

A healthcare organization wants to develop a deep learning system to automatically classify chest X-ray images into **normal** and **disease-affected** categories. The dataset contains thousands of X-ray images with variations in size, brightness and image quality. A CNN model is proposed because it can automatically learn important visual features such as edges, textures, and abnormal regions without manually designing features.

Question:

1. Design and explain a suitable **CNN architecture** for this medical image classification system.
2. Describe the **motivation for using CNNs** and explain the role of **convolutional layers, filters, pooling layers, and fully connected layers** in the proposed architecture.
3. Explain how **parameter sharing** reduces the number of trainable parameters compared with a fully connected network.
4. Discuss suitable **regularization techniques such as dropout, data augmentation and batch normalization** to reduce overfitting.
5. Finally, explain whether **AlexNet or ResNet** would be more appropriate for this application and justify your choice based on feature learning, network depth, computational requirements and classification performance.

Answer:**1. Suitable CNN Architecture for Medical Image Classification**

A CNN model for chest X-ray classification can be designed as follows:

Input Layer

- Input image size: $224 \times 224 \times 3$

Convolution Layer 1

- 32 filters (3×3)
- ReLU activation

Max Pooling Layer 1

- Pool size: 2×2

Convolution Layer 2

- 64 filters (3×3)
- ReLU activation

Max Pooling Layer 2

- Pool size: 2×2

Convolution Layer 3

- 128 filters (3×3)
- ReLU activation

Max Pooling Layer 3

- Pool size: 2×2

Flatten Layer**Fully Connected Layer**

- 128 neurons
- ReLU activation

Dropout Layer

- Dropout rate: 0.5

Output Layer

- 2 neurons (Normal / Disease)

- Softmax activation

Working

The model receives chest X-ray images and automatically extracts important features such as edges, lung textures, and abnormal regions. These features are then used to classify images into normal or disease-affected categories.

2. Motivation for Using CNNs and Role of Layers

Why CNN?

CNNs are preferred because they:

- Automatically learn image features.
- Require less manual feature engineering.
- Achieve high accuracy in medical image analysis.
- Handle image variations effectively.

Role of Components

Convolutional Layer

- Extracts features from images.
- Detects edges, textures, and shapes.

Filters

- Small matrices that scan images.
- Learn disease-specific patterns automatically.

Pooling Layer

- Reduces image dimensions.
- Decreases computation.
- Retains important information.

Fully Connected Layer

- Combines extracted features.
- Performs final classification.

3. Parameter Sharing in CNN

In a Fully Connected Network:

If input image size is:

$$224 \times 224 = 50,176 \text{ pixels}$$

Connecting to 100 neurons:

Parameters =

$$50,176 \times 100$$

$$= 5,017,600 \text{ parameters}$$

In CNN:

Suppose 32 filters of size 3×3 are used.

Parameters =

$$(3 \times 3 \times 1 + 1) \times 32$$

$$= 320 \text{ parameters}$$

Advantage

- Same filter is used across the entire image.
- Greatly reduces memory usage.
- Faster training.
- Better generalization.

This concept is called **Parameter Sharing**.

4. Regularization Techniques to Reduce Overfitting

A. Dropout

- Randomly deactivates neurons during training.
- Prevents dependency on specific neurons.
- Improves generalization.

Example:

- Dropout rate = 0.5

B. Data Augmentation

Artificially increases dataset size.

Techniques:

- Rotation
- Flipping
- Zooming
- Brightness adjustment
- Cropping

Benefits:

- Improves robustness.
- Reduces overfitting.

C. Batch Normalization

- Normalizes activations.
- Stabilizes training.
- Enables faster convergence.
- Reduces internal covariate shift.

5. AlexNet vs ResNet for Medical Image Classification

Feature	AlexNet	ResNet
Year	2012	2015
Depth	8 Layers	50, 101, 152+ Layers
Accuracy	Good	Very High
Feature Learning	Moderate	Excellent
Vanishing Gradient	Possible	Solved using Skip Connections
Medical Image Performance	Good	Better
Computational Cost	Lower	Higher

Recommended Model: ResNet

Justification

- Learns deeper features.
- Detects subtle abnormalities in X-rays.
- Skip connections avoid vanishing gradients.
- Higher classification accuracy.
- Widely used in medical imaging applications.

Conclusion

ResNet is more suitable because medical images require deep feature extraction and high diagnostic accuracy.

Case Study Question 2: CNN-Based Autonomous Vehicle Object Recognition (10 Marks)

An autonomous vehicle company is developing a vision system that must identify **cars, pedestrians, traffic signs, bicycles, and road obstacles** from images captured by cameras mounted on the vehicle. The system must recognize objects under different lighting conditions, viewing angles, distances and road environments. The development team is considering **AlexNet and ResNet** architectures for building the CNN-based recognition system.

Question:

1. Analyze how a **CNN architecture** can be designed for this autonomous vehicle application.
2. Explain the purpose of the **input, convolution, activation, pooling, and fully connected/output layers** and describe how convolutional **filters progressively learn low-level and high-level features**.
3. Explain the importance of **parameter sharing** in reducing computational complexity and improving translation-related feature detection.
4. Discuss the possible problem of **overfitting** and recommend suitable regularization methods.
5. Compare **AlexNet and ResNet** in terms of architecture, depth, parameter efficiency, training difficulty, vanishing-gradient problems and feature-learning capability.
6. Finally, recommend the most suitable architecture for the autonomous vehicle system and justify your decision based on **accuracy, computational cost and real-time application requirements**.

Answer:

1. CNN Architecture for Autonomous Vehicles

Input Layer

- Camera image ($224 \times 224 \times 3$)

Convolution Layer 1

- 32 filters (3×3)
- ReLU

Max Pooling Layer

Convolution Layer 2

- 64 filters (3×3)
- ReLU

Max Pooling Layer

Convolution Layer 3

- 128 filters (3×3)
- ReLU

Max Pooling Layer

Flatten Layer

Fully Connected Layer

- 256 neurons

Output Layer

- Classes:
 - Car
 - Pedestrian
 - Bicycle
 - Traffic Sign
 - Obstacle

Softmax Activation

The CNN identifies objects under different lighting conditions, distances, and viewing angles.

2. Purpose of Layers and Feature Learning

Input Layer

Receives vehicle camera images.

Convolution Layer

Extracts image features.

Activation Layer (ReLU)

Introduces non-linearity.

Formula:

$$\text{ReLU}(x) = \max(0, x)$$

Pooling Layer

Reduces dimensions and computation.

Fully Connected Layer

Performs classification.

Progressive Feature Learning

Early Layers

- Edges
- Corners
- Lines

Middle Layers

- Wheels
- Windows
- Human body parts

Deep Layers

- Cars
- Pedestrians
- Traffic signs

3. Importance of Parameter Sharing

Parameter sharing means the same filter is applied across the image.

Benefits

- Fewer trainable parameters.
- Reduced memory usage.
- Faster computation.
- Detects objects regardless of position.

Example:

A pedestrian appearing on the left or right side can still be recognized using the same filter.

4. Overfitting and Regularization Methods

Dropout

Randomly removes neurons during training.

Data Augmentation

- Rotation
- Brightness variation
- Scaling
- Cropping

Batch Normalization

Improves stability and convergence.

Early Stopping

Stops training when validation loss increases.

These techniques improve real-world performance.

5. AlexNet vs ResNet Comparison

Feature	AlexNet	ResNet
Architecture	Sequential CNN	Residual CNN
Layers	8	50+
Parameters	More	More Efficient
Training Difficulty	Moderate	Easier due to Skip Connections
Vanishing Gradient	Present	Solved
Feature Learning	Good	Excellent
Accuracy	High	Very High
Object Recognition	Good	Better

Residual Block

Output:

$$F(x) + x$$

where:

- $F(x)$ = learned mapping
- x = shortcut connection

This enables deep network training.

6. Recommended Architecture

Recommended: ResNet

Reasons

1. Higher recognition accuracy.
2. Better feature extraction.
3. Handles complex road scenes.
4. Solves vanishing gradient problem.
5. Detects small and distant objects effectively.
6. Performs well in autonomous driving applications.

Conclusion

ResNet is the best choice for autonomous vehicle object recognition because it provides superior accuracy, deep feature learning, and robust performance under varying road and environmental conditions.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand
